bioelectric potential on the surface of the human body such as Electrocardiogram (ECG), Electroencephalograph (EEG), Electromyogram (EMG), Electroculogram (EOG), etc., play a vital role in understanding the functioning of the body organs and their disease condition. In automobiles, most of the mechanical parts are controlled by electronic controls. Additionally, the identification of human beings is based on biometrics measurements, which completely depends on electronic gadgets. As a result, the human life is dominated by electrical and electronic gadgets, hence, the learning and understanding of their working principle is well advised for proper handling, maintenance and troubleshooting of those devices. Hence, we will discuss the history and basic principles of electricity in this Chapter.

1.1.1 History of Electricity

In June 1752, American Scientist Benjamin Franklin, tried to prove that lightning is a phenomenon of electricity by flying a kite with conductive rods to attract lightning to a Leyden jar, which was an early form of a capacitor. When kite was flying at a certain height, he observed an electric spark and felt electric shock slightly, which proved that lightning is electric in nature.

In 1800, in Paris (France), Italian scientist Alessandro Volta proved that his invention, "Voltaic Pile" produced electricity.



Later days, The Voltaic Pile was called as Voltaic cell

1.1.2 Generation of Electricity

In 1831-32, a British scientist Michael Faraday invented electromagnetic generator using Faraday's laws of electro magnetism which produced Alternating current. Nikola Tesla, Serbian-American inventor contributed the design of modern A.C supply system.

1.1.2 (a) Types of Power Generation

Important types of power generation are

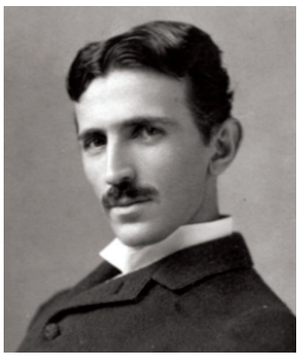
1. Hydroelectric power station
2. Thermal power station
3. Atomic power station
4. Wind Power station
5. Solar power station (D.C)
6. Gas power station



Benjamin Franklin



Benjamin Franklin Trying to Draw Electricity from Lightning



Nikola Tesla

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